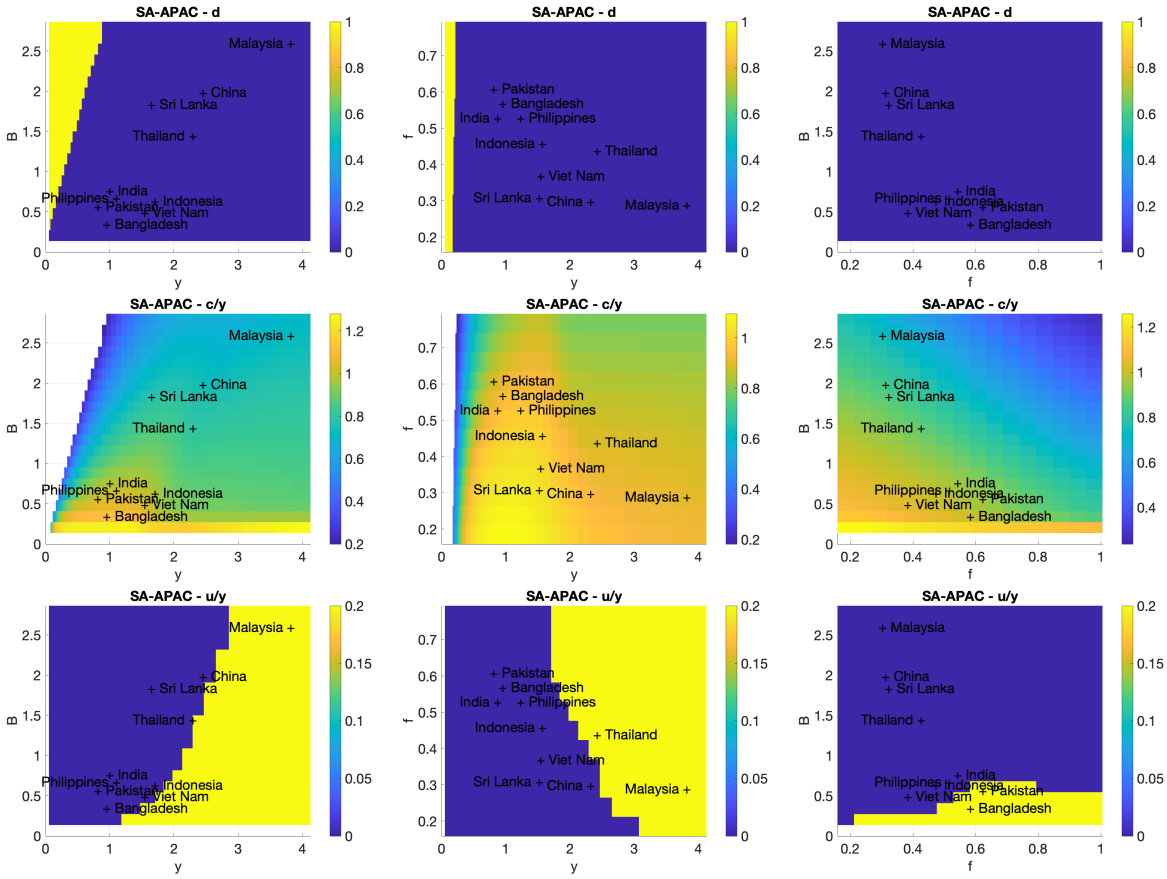

Figure 4: SA-APAC region: decision to default (d), consumption over output (c/y) adaptation expenditure over output (u/y). When a variable is fixed, it is set equal to the median value for the macro-region, see Table 7: plane (y, B), $f = 0.45$; plane (y, f), $B = 0.72$; plane (f, B), $y = 1.54$. The white region in the picture for c/y corresponds to the default region. The positioning of the country labels in the pictures is intended to be indicative because, for example, in the (y, B) plane, the values of d, c/y, and u/y correspond to the median value of f (specifically, $f = 0.45$).

macro-regions. The relationship between consumption over output and output is bell shaped for most of the values of the state variables B and f in the the SA-APAC and SSA macro-region and is increasing in the LATAM macro-region. As far as dependence on public debt is concerned, we observe a negative relationship in all three macro-regions.

As far as adaptation expenditure is concerned, we observe very little evidence that countries take action to adapt to climate risks. Positive investment to address climate risks is observed only when $y \gg B$ or $y \gg f$ for the SA-APAC and the SSA macro-region, while in the LATAM macro-region there is little space for adaptation expenditure. This outcome is due to the fact that, as already observed, LATAM countries are less exposed to environmental risk. An increase in f reduces the activation threshold for y, that is, the minimum level of y at which a country begins to take action on climate change. This effect is more pronounced in the SA-APAC macro-region compared to the SSA macro-region.

Note that there is no country in the LATAM region that should invest in climate adaptation (u/y is always 0, see Table 6, in line with Figure 3), instead there are some countries with positive investment in the SSA and SA-APAC macro-region (Botswana and Mauritius, see Table 5, and Malaysia, see Table 7), the main driver for the decision being the high values of y, as also shown in Figure 2 and 4.

7 Sensitivity analysis and policy options

In Appendix C we provide a sensitivity analysis with respect to the main parameters of the model: χ, η, k_f, λ, σ_f, θ_d, φ, g, ν_0. We perform the analysis showing the difference in d, c/y, u/y between the value obtained for the median parameter augmented by 20% and the value obtained decreasing it by 20%. We provide the analysis for the SA-APAC macro-region, results for the other regions are similar and available upon request.

Increasing χ (decreasing the average duration of exclusion from issuing public debt) does not affect neither the decision to default on debt nor to abate pollution, see Figure C.1. A very low positive effect is observed for the consumption ratio.

Increasing the parameter η (the fraction of the output recovered after the default event), the region where it is optimal to default increases, as expected, while the region where it is optimal to have a non null adaptation expenditure decreases, see Figure C.2. It seems that a higher recovering rate leads the country to care less of climate risk. Mild renegotiation conditions lead to less action against climate risk. The effect on the

consumption ratio is negative.

As the amount of pollution removed per unit of expenditure (k_f) increases, the region with a positive adaptation expenditure significantly increases, see Figure C.3. No effect is observed on the default decision, the effect on the consumption ratio is mixed. It is positive in general, but it becomes negative when climate risk action becomes positive.

The default region slightly increases as λ increases (debt maturity becomes shorter), see Figure C.4. No effect is observed on the adaptation expenditure decision, the effect on the consumption ratio is negative the rationale being that short maturity doesn't provide hedge to output fluctuations, see [6].

As the debt fraction after default θ_D increases, no effect is detected on the default and risk adaptation expenditure decision, see Figure C.6, the effect of the consumption ratio is limited.

The parameter φ models how the pollution decays by natural sinks. As this parameter increases, the region with a positive adaptation expenditure shrinks with a positive effect on the consumption ratio, no effect on the default decision is observed, see Figure C.7.

Several parameters do not affect the outcome of the model: uncertainty surrounding pollution evolution, technology improvement ($\sigma_f,\ g$), see Figure C.5, C.8. The baseline frequency for a climate disaster ν_0 has no effect on the default and adaptation risk expenditure, the effect on the consumption ratio is negative but rather small, see Figure C.9. This confirms the weak nexus between debt management and climate risk.

We consider two policy options to promote adaptation risk expenditure: carbon offset incentive and readmission to financial markets dependent on pollution reduction.

Thanks to a carbon offset incentive, the country is reimbursed a certain fraction of the expenditure to decrease the stock of pollution. To model this incentive we add term $-pu(t)$ in (8), that is, the budget constraint becomes:

$$Q(t)B^{new}(t) = (\lambda + \delta)B(t) + c(t) + (1 - p)u(t) - y(t).$$

The idea comes from the UN REDD+ initiative, which is a global program developed by the United Nations to combat climate change by reducing emissions from deforestation and forest degradation in developing countries[2].

We investigate this option assuming a 20% incentive, i.e., $p = 0.2$. Results are reported

[2]https://unfccc.int/topics/land-use/workstreams/redd/what-is-redd.